Phase 8 — Disaster Recovery Test and Executive Report

Harrington Capital plc | Infrastructure Engineer Simulation

Objective

Execute and document a DORA-compliant DR test. Verify backup restore integrity. Write the executive infrastructure summary for Sophie Cartwright (CTO).

DORA Article 11 and Article 26 require annual DR testing with documented evidence. The last test was 14 months ago. This is overdue.

By the end of this phase you will have: - A backup created and restored — with evidence - DR runbook committed to `/runbooks` - Executive report (1 page) committed to `/evidence` - All evidence uploaded and committed

---

BUILD

Task A — Create and Restore a Backup

Step 1 — Create a backup of HC-APP01 data

On HC-APP01:

```
# Create backup directory
sudo mkdir -p /var/backups/harrington

# Backup the Docker volumes
docker compose -f ~/harrington-infrastructure-operations/docker/docker-compose.yml down

# Export the database
docker run --rm \
    -v hc-db-data:/data \
    -v /var/backups/harrington:/backup \
    ubuntu \
    tar czf /backup/hc-db-$(date +%Y%m%d-%H%M).tar.gz -C /data .

# Record the backup file
ls -lh /var/backups/harrington/
```

Bring the stack back up:

```
docker compose -f ~/harrington-infrastructure-operations/docker/docker-compose.yml up -d
```

Step 2 — Simulate a failure

```
# Stop the database container to simulate failure
docker stop hc-db
docker rm hc-db

# Verify the app is returning errors (as expected)
curl -s http://localhost:8080/db-check
# Expected: error response
```

Take a screenshot showing the failed state.

Step 3 — Restore from backup

```
# Create a new empty volume
docker volume create hc-db-data-restored

# Restore the backup
BACKUP_FILE=$(ls /var/backups/harrington/hc-db-*.tar.gz | tail -1)
docker run --rm \
    -v hc-db-data-restored:/data \
    -v /var/backups/harrington:/backup \
    ubuntu \
    tar xzf /backup/$(basename $BACKUP_FILE) -C /data

echo "Restored from: $BACKUP_FILE"

# Update docker-compose to use restored volume and bring stack up
docker compose -f ~/harrington-infrastructure-operations/docker/docker-compose.yml up -d
```

Step 4 — Verify restoration

```
curl -s http://localhost:8080/health
curl -s http://localhost:8080/db-check
```

Both must return healthy responses.

```
df -h
docker ps
```

Document the Recovery Time: time from `docker stop` to application healthy again.

Task B — Complete the DR Runbook

Copy `/runbooks/RB-DR-v1.md` and complete all sections with your actual findings.

Key metrics to record:

Metric	Your Value
Backup size	e.g. 142 MB
Backup creation time	e.g. 47 seconds
Restore time	e.g. 2 minutes 18 seconds
RTO achieved	e.g. 8 minutes total
RPO (data age)	e.g. same-day backup

Task C — Executive Report (1 page)

Write the executive infrastructure summary for Sophie Cartwright (CTO).

Save as `/evidence/phase-8-exec-report.md`.

Format rules: - Maximum 1 page (400 words) - No jargon - No tool names unless essential - Risk status: what was bad, what is now fixed - Numbers only — no vague statements

Required sections:

```
HARRINGTON CAPITAL — INFRASTRUCTURE STATUS REPORT
Date: [today]
Prepared by: [your name]
Recipient: Sophie Cartwright, CTO

EXECUTIVE SUMMARY (3 sentences max)
What the situation was, what you did, what the risk status is now.

COMPLETED REMEDIATION (table)
Item | Previous State | Current State | DORA Obligation Met?
Infrastructure IaC | Manual portal only | Terraform committed | —
AD documentation | None | OU structure documented, 10 accounts | —
Monitoring | 11 hosts not covered | All hosts in Prometheus/Grafana | Art. 17
Incident process | No formal RCA filed | GLPI operational, RCA filed | Art. 17
DR test | 14 months overdue | Test completed, evidence filed | Art. 11 + Art. 26
Automation | Manual operations | PowerShell + Ansible suite committed | —

CURRENT RISK STATUS (RAG)
Red items: [none / list any]
Amber items: [list outstanding items]
Green items: [list completed]

COST IMPACT
Estimated time saved per month by automation: X hours
Estimated risk reduction: [qualitative]

OUTSTANDING ACTIONS
Item | Owner | Due date
```

VERIFY

```
# Backup file exists and has size
ls -lh /var/backups/harrington/

# Application healthy after restore
curl -s http://localhost:8080/health | python3 -m json.tool

# DB check passes
curl -s http://localhost:8080/db-check | python3 -m json.tool
```

SUBMIT

#	Evidence	How to get it
1	Backup file screenshot	`ls -lh /var/backups/harrington/` showing file with date and size
2	Failed state screenshot	`curl /db-check` returning error during simulated failure
3	Restored state screenshot	`curl /health` and `curl /db-check` returning healthy
4	DR runbook GitHub URL	`/runbooks/RB-DR-v1.md` committed
5	Executive report GitHub URL	`/evidence/phase-8-exec-report.md` committed

Anti-fake check: Backup filename must contain today's date (YYYYMMDD format). Restore time must be stated in minutes and seconds — not estimated.